

SUMMARY

Senior Systems Architect owning the backend platform and cloud infrastructure of a smart-home IoT product. Took over a legacy, manually deployed stack and turned it into a stable, high-ingestion platform: refactored legacy services, removed recurring outage causes at the root, and made zero-downtime migrations and controlled rollouts the default. Own the custom binary UDP transport the device fleet speaks today and drive R&D on its next-generation successor. Author the protocol specs and data contracts that firmware, mobile app, and backend teams build against, and propose architectural changes across those teams.

TECHNICAL COMPETENCIES

|                       |                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LANGUAGES             | Kotlin, Java, Go, Rust, Zig, C, Shell scripting (POSIX/Bash), SQL                                                                                                                                                |
| KERNEL & NETWORKING   | Linux kernel ( <code>nftables</code> , <code>iptables</code> ), in-kernel NAT64 ( <code>Jool</code> ), TCP/UDP socket programming, mTLS & custom PKI (CA/CRL), packet reflection, conntrack tuning               |
| DISTRIBUTED & STORAGE | ClickHouse (columnar modeling, Delta+ZSTD codecs), Redis, PostgreSQL, MongoDB, gRPC, Protocol Buffers, REST                                                                                                      |
| CLOUD & SRE           | Google Cloud (IAM, VPC, Cloud Run, Cloud SQL, GCS), AWS (Lambda, EKS spike), Terraform / OpenTofu IaC, Docker, Proxmox VE, GitHub Actions, Linux administration (Debian, Ubuntu)                                 |
| OBSERVABILITY         | Grafana, Loki, VictoriaMetrics, VictoriaLogs, Alloy, <code>async-profiler</code> , JMC, VisualVM, Linux performance tools ( <code>sar</code> , <code>vmstat</code> , <code>iotop</code> , <code>pidstat</code> ) |
| METHODOLOGY           | Extreme Programming (XP), London-Style TDD, Consumer-Driven Contract Testing (Pact spike), Testcontainers                                                                                                        |

PROFESSIONAL EXPERIENCE

eGlu Smart Homes (WiZSystems)

Bengaluru, India · 2022 – Present

Senior Systems Architect (Apr 2026 – Present) | Senior Software Engineer (2024 – 2026) | Software Engineer (2022 – 2024)

IoT Platform & High-Ingestion Protocols

- Custom UDP Transport for the Device Fleet:** Define and maintain the custom binary UDP protocol that smart-home hubs use to talk to the cloud, and the high-ingestion backend that decodes it. Tuned the ingest path end to end for throughput and low per-packet latency, from kernel-level packet handling to a load-adaptive JVM thread pool and allocation-free packet decoding.
- Next-Generation Protocol R&D:** Driving R&D on the next-generation device transport protocol to succeed the current UDP transport.
- Cross-Team Protocol & Architecture Proposals:** Author the binary protocol definitions, API contracts, and technical specs (Git-based knowledge base) that firmware, mobile app, and backend teams review and implement against before releases. Propose architectural changes to the firmware and app teams, including the SNode virtual-node abstraction (aggregating disjoint physical devices into one composite device) and backward-compatible mobile rollout contracts.

Legacy Revamp & Platform Stabilization

- Legacy Platform Modernization:** Took full technical ownership of the backend and cloud infrastructure in 2023. Replaced manual SFTP WAR deployments with automated CI/CD, migrated Spring 4 (JDK 8) to Spring Boot 2.7.5 (JDK 17) to unlock TDD, rewrote the device provisioning and WiFi onboarding state machine, and removed obsolete dynamic TCP port provisioning and redundant in-memory state tracking.
- Stabilizing Production:** Eliminated recurring outages at the root: systemd file-descriptor limits starving thread allocation and locking out SSH, an auxiliary service leaking unpooled MongoDB connections, and silent "ghost connections" poisoning Redis Pub/Sub pools (Lettuce vs. Jedis). Fixed a months-old legacy sync bug silently corrupting hub automations and reconciled state across thousands of live homes without breaking customer automations.
- Reducing Downtime with Deliberate Release Strategy:** Made zero-downtime the default: kernel-level blue-green cutovers ( `iptables` DNAT across PREROUTING and OUTPUT , `conntrack` flush), atomic and idempotent deploy tooling with pre-flight health checks and rollback guards after a live incident, a mandatory on-prem staging gate mirroring production topology, and a feature flags engine (percentage rollouts, instant kill switches) for high-blast-radius cloud, mobile, and firmware changes.

Systems Architecture & Kernel Networking

- In-Kernel Dual-Stack NAT64 ( `Jool` + `nftables` ):** Resolved an edge network partition where cellular IoT hubs operated strictly over IPv6 while backend services were IPv4. Bypassed userspace proxies to preserve CPU credits; deployed the `Jool` kernel module for stateful NAT64 alongside `nftables` DNAT to cleanly partition source port ranges on a shared public IPv4.
- Prerouting Packet Reflection & Dynamic Abuse Mitigation:** Offloaded UDP NAT traversal echo responses directly into the Linux kernel using `nftables` packet reflection ( `notrack` ) at prerouting priority, avoiding userspace context switches. Implemented kernel-level DDoS mitigation via `nftables` dynamic sets ( `limit rate over 10/second burst 20 packets` at prerouting priority -301) dropping sweeps before socket allocation.
- Academic Preprint Publication (Zenodo):** Authored and published *Connection-Agnostic Presence Tracking for Stateless Distributed Backends*: formulated a Redis sorted-set architecture with expiration deadlines and throttled batch writes to track IoT device state with mathematically bounded detection latency.

## High-Throughput Telemetry & Storage Systems

- **ClickHouse Ingestion Engine (95%+ Compression):** Resolved filesystem inode exhaustion caused by legacy directory log dumps. Designed high-throughput columnar ClickHouse schemas ( `FixedString` , `LowCardinality` , Delta + ZSTD codecs, sparse composite primary keys), slashing telemetry footprint by 95%+ (compressing `device_health` telemetry from 17 GB down to 300 MB on disk). Executed zero-downtime partition-level backfilling and restore operations without slow `INSERT INTO` batches, and devised partition-level disaster recovery backup and restore policies.
- **Decoupled Ingestion Microservice:** Engineered an auxiliary Go microservice communicating via binary gRPC with strict Protocol Buffers contracts to encapsulate ClickHouse, shielding the primary Kotlin/Spring Boot monolith from database driver coupling.
- **Zero-Downtime Storage Architecture:** Architected and orchestrated an end-to-end multi-phase data migration decoupling binary Base64 image blobs from primary MongoDB collections to GCS with zero downtime; designed dual-write API contracts with document guardrails ( `imageMigrated` ), executed an asynchronous backfill pipeline with immutable GCS caching headers, coordinated backward-compatible mobile rollouts, and purged legacy blobs via `$unset` , slashing database storage by 54x (3.6 GB to 67 MB) and runtime memory by 89% (4.6 GB to 500 MB).
- **Production MongoDB DBA & Index Engineering:** Acted as hands-on DBA across heavy-load production MongoDB clusters. Audited slow query logs and execution plans ( `explain` ), pruned redundant/overlapping indexes to reclaim WiredTiger cache memory and reduce write amplification, refactored hot document schemas, and engineered selective compound indexes (strict Equality-Sort-Range ordering) to eliminate collection scans and in-memory sorts under sustained IoT write traffic.

## Concurrency, Runtime & JVM Performance

- **Dynamic IoT-Aware ThreadPool:** Overhauled JVM thread management under bursty IoT packet waves by replacing default thread pools with an adaptive dynamic ThreadPool that scales worker threads based on real-time packet velocity and queue depth.
- **Deep JVM Profiling & Zero-Allocation Hot Paths:** Instrumented live production systems with `async-profiler` and JMC flame graphs. Mitigated allocation churn with buffer pooling and `ThreadLocal` storage; refactored critical packet decoding loops from `String.format` to Java 17 `HexFormat` and bitwise arithmetic, eliminating hot-path CPU bottlenecks. Pinned heap boundaries ( `-Xms = -Xmx` at 4GB) and tuned GC for low pause times.

## Platform Engineering, Governance & Cloud FinOps

- **Google Cloud Platform Administration & FinOps:** Google Cloud Platform Administrator managing IAM policies, VPC networks, Cloud Storage, and compute infrastructure. Evaluated spend-based vs. resource-based CUD models and executed a 3-year resource-based Committed Use Discount (CUD) on core compute instances.
- **Observability & Telemetry Infrastructure:** Replaced ad-hoc raw log dumping on production VM disks with centralized log aggregation using Grafana and Loki. Built a pre-production Docker Compose telemetry stack integrating Grafana, Loki, VictoriaMetrics, VictoriaLogs, and Alloy with host and container performance dashboards.
- **Zero-Trust Diagnostic Tooling ( `hlogger` ):** Built a secure client-server log inspection utility in Go using Mutual TLS (mTLS) with a custom PKI hierarchy (Root CA, Intermediate CA, client certificates, CRL), eliminating unauthenticated engineer SSH access to production instances.
- **Bare-Metal Homelab:** Configured a Proxmox VE bare-metal host provisioned via OpenTofu and `cloud-init` to evaluate container orchestration (k3s, HashiCorp Nomad, AWS EKS spikes).
- **Declarative Marketing Infrastructure:** Decoupled the public marketing website completely from core backend infrastructure into its own isolated GCP project, codifying 100% of Cloud Run, Cloud SQL (PostgreSQL), networking, and IAM via Terraform / OpenTofu.
- **Platform Automation & Integrations:** Extended scene/rule automation engines for multi-hub environments (fragment handling, cross-hub synchronization); built cloud-to-cloud OAuth integrations for Yale smart locks, Google Home, and Alexa; built replacement operations for failed hardware nodes without losing room mappings or automations.
- **Technical Hiring:** Formulated structured, multi-step technical hiring assessments and evaluation rubrics for backend engineering; interviewed candidates and hired software engineering talent for the platform.

## OPEN-SOURCE SYSTEMS & RESEARCH

---

**stopgap** Kotlin · Maven Central ( 2.8.0 )

Microservice framework on Helidon SE (Nima) + Project Loom virtual threads. Custom compile-time DI via KSP (zero runtime reflection). Three-tier testing harness: unit → in-process integration server → Docker E2E via Testcontainers.

**assertgo** Go · Generics

Type-safe testing assertion library built with modern Go generics. Provides a fluent assertion API, chainable `Not()` , custom matchers, and zero external dependencies.

**relay** Zig · Systems

Low-level TCP server in pure Zig with a test-driven approach: raw POSIX socket descriptors, manual memory management without libc runtime dependencies, port binding ( `SO_REUSEADDR` ), and `SIGPIPE` suppression.

**hrh** Rust · CLI

Helm Release Helper CLI — automated declarative Helm deployments with diff previews and atomic rollback guarantees via `cargo install` .

**avoid** Shell / Linux · Distro

Minimal, bootable Linux distribution based on Void Linux engineered for server disaster recovery and lightweight headless appliances.

## EDUCATION

---

**Bachelor of Computer Applications (BCA)**

2019 – 2022